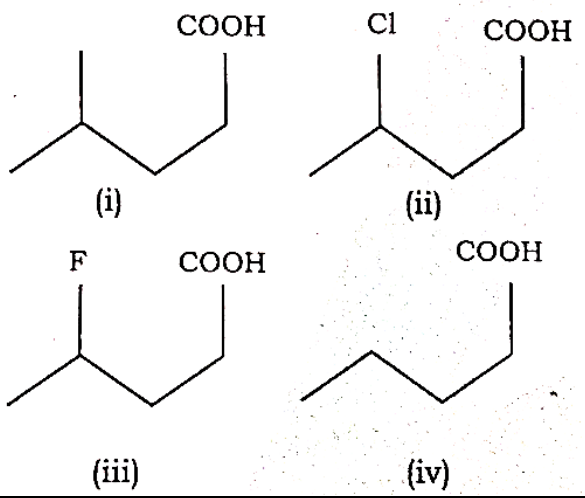
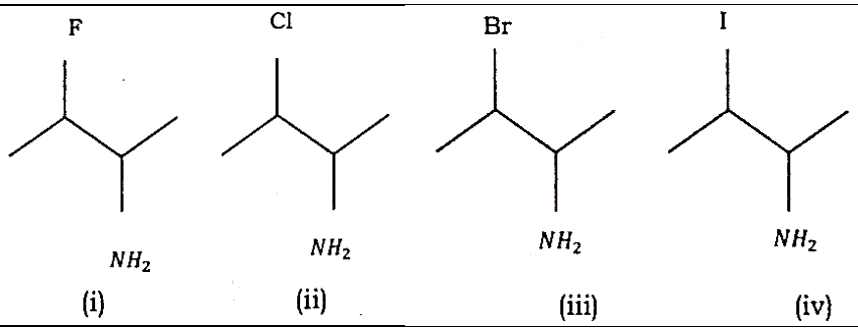
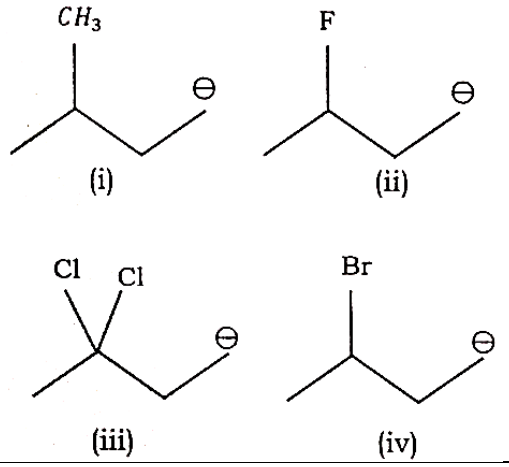
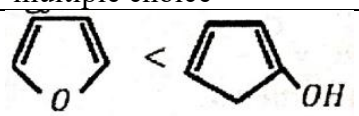
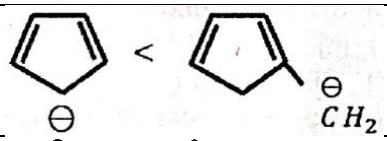
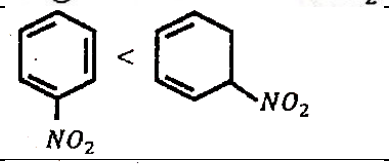
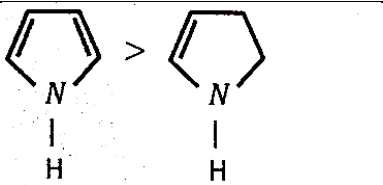
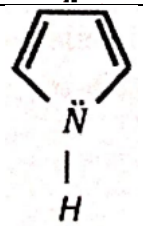
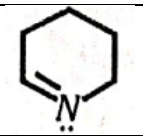
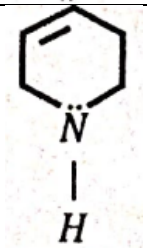
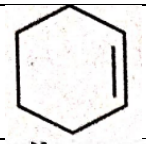
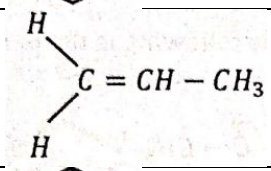
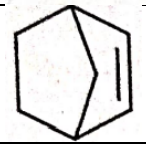
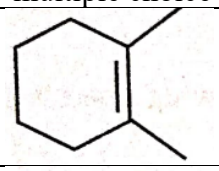
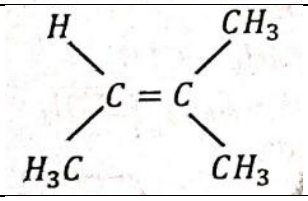
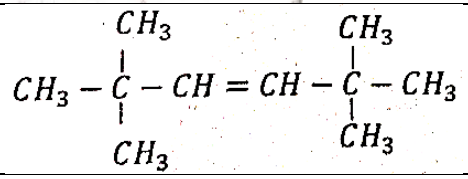
Question	Which one of the following orders is correct regarding the inductive effect of the substituents?
Type	multiple choice
Option	$-NR_2 < -OR > -F$
Option	-NR2> -OR > -F
Option	-NR2 < −OR < -F
Option	-NR2 > OR < -F
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Basic strength of (i) $H_3C\overset{\ominus}{C}H_2$ (ii) $H_2C=\overset{\ominus}{C}H$ (iii) $H-C=\overset{\ominus}{C}$ Is in the order of
Type	multiple choice
Option	(i) > (iii) > (ii)
Option	(i) > (ii) > (iii)
Option	(ii)> (i) > (iii)
Option	(iii) > (ii)> (i)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct order of acidic strength of the following compounds is: 
Type	multiple choice
Option	(iii) > (i) > (iv) > (ii)
Option	(iii))> (ii) > (iv) > (i)
Option	(iv) > (ii) > (i) > (iii)
Option	(i) (ii) > (iii) > (iv)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct order of basic strength of the following compounds is:

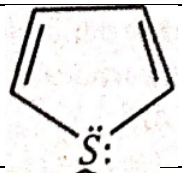
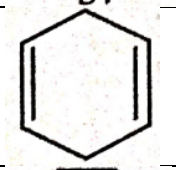
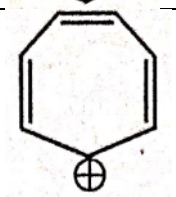
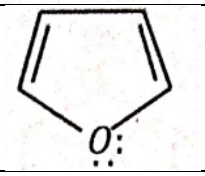
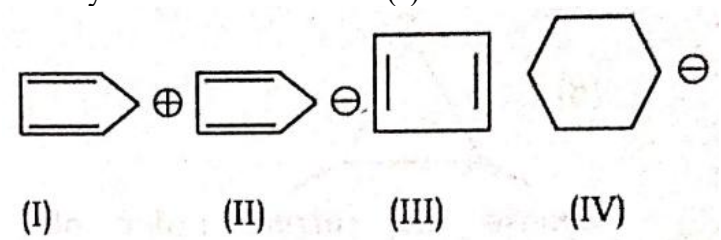
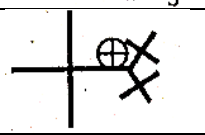
	
Type	multiple choice
Option	(i) > (ii) > (iii)> (iv)
Option	(ii) > (iii) > (iv) > (i)
Option	(iv) > (ii) > (i) > (iii)
Option	(iv) > (iii) > (ii) > (i)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The decreasing order of basic strengths of the following compounds is: 
Type	multiple choice
Option	(i) < (iii) > (ii) > (iv)
Option	(iii) > (ii) > (iv) > (i)
Option	(iv) > (ii) > (i) > (iii)
Option	(i) > (iv) > (ii) > (iii)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Most stable carbocation is:
Type	multiple choice
Option	$CH_3 - CH_2 - \overset{\oplus}{C}H_2$
Option	$Br - CH_2 - CH_2 - \overset{\oplus}{C}H_2$
Option	$Cl - CH_2 - CH_2 - \overset{\oplus}{C}H_2$
Option	$F - CH_2 - CH_2 - \overset{\oplus}{C}H_2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

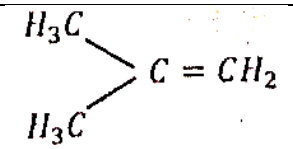
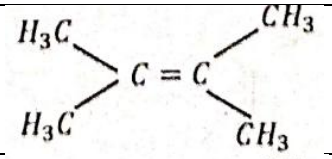
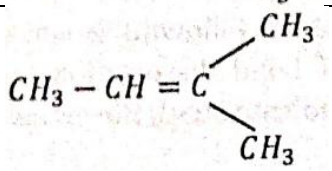
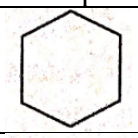
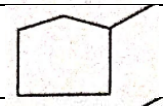
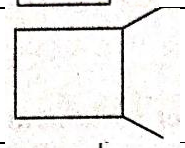
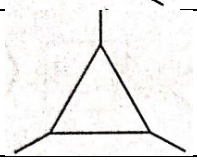
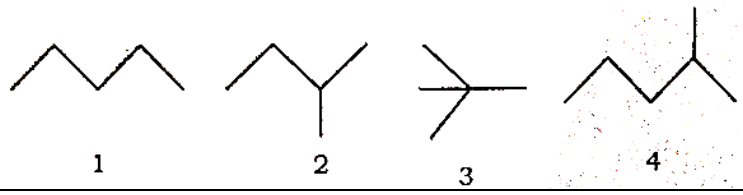
Question	Most stable carbanion is:
Type	multiple choice
Option	$\overset{\ominus}{C}H_2 - CH_2 - NO_2$
Option	$\overset{\ominus}{C}H_2 - CH_2 - CN$
Option	$\overset{\ominus}{C}H_2 - CH_2 - COOH$
Option	$\overset{\ominus}{C}H_2 - CH_2 - CH_2 - CH_2 - F$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is most acidic:
Type	multiple choice
Option	$Cl_3C - COOH$
Option	$Cl_2CH - COOH$
Option	$ClCH_2 - COOH$
Option	CH_3COOH
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is most acidic:
Type	multiple choice
Option	$\underset{\underset{F}{\vert}}{C}H_2 - COOH$
Option	$\underset{\underset{Cl}{\vert}}{C}H_2 - COOH$
Option	$\underset{\underset{Br}{\vert}}{C}H_2 - COOH$
Option	$\underset{\underset{I}{\vert}}{C}H_2 - COOH$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is most basic among the following:
Type	multiple choice
Option	$CH_2CH_2 - NH_2$
Option	$Cl - CH_2CH_2NH_2$
Option	$O_2N - CH_2CH_2 - NH_2$
Option	$^{-}O - CH_2CH_2 - NH_2$
Answer	4
Solution	

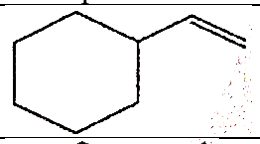
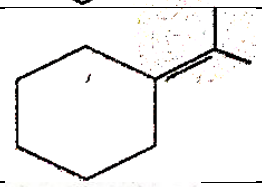
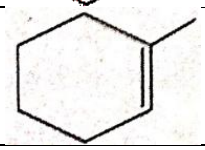
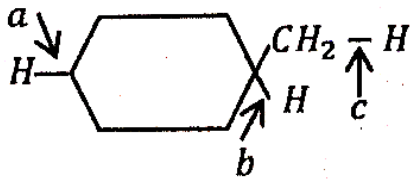
Positive Marks	4
Negative Marks	1
Question	The decreasing order of stability of the following resonating structures is $CH_2 = CH - \ddot{\underset{..}{C}}l: \leftrightarrow : \overset{\ominus}{C}H_2 - CH = \overset{\oplus}{\underset{..}{C}}l: \leftrightarrow$ $\overset{\oplus}{C}H_2 - CH = \overset{\ominus}{\underset{..}{\ddot{C}}}l:$
Type	multiple choice
Option	I > II > III
Option	II > III > 1
Option	III > II > I
Option	I > III > II
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following pairs represent resonance?
Type	multiple choice
Option	$CH_2 = CHOH\,;CH_3CHO$
Option	$\overset{\ominus}{C}H_2 - CHO; CH_2 = CH - \overset{\ominus}{O}$
Option	$CH_3 - \overset{O}{\overset{\|\|}{C}} - CH_3; CH_3 - \overset{OH}{\overset{\|}{C}} = CH_2$
Option	$CH_3 - \overset{OH}{\overset{\|\|}{C}} - CH_3; CH_3 - \overset{OH}{\overset{\|}{C}} = CH_2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is correctly ordered for resonance stability?
Type	multiple choice
Option	$H_2\overset{\ominus}{C} - CH = \overset{\oplus}{N} = O \leftrightarrow$ (I) $CH_2 = CH - \ddot{N} = O \leftrightarrow \overset{\oplus}{C}H_2 - CH = N - O^{\ominus}$ (II) (III) (II > I > III)
Option	$\overset{\oplus}{N}H_2 = C = O \leftrightarrow NH_2 - \overset{\oplus}{C} = O \leftrightarrow NH_2 - C \equiv \overset{\oplus}{O}$ (I) (II) (III) (I > III > II)
Option	$H_2C - \overset{\oplus}{C} = O \leftrightarrow H_3C - C \equiv \overset{\oplus}{O}$ (I) (II) (I > II)
Option	$\overset{\oplus}{C}H_2 - \overset{\ominus}{C}H - \overset{\oplus}{N} \begin{matrix} O \\ // \\ \diagdown O^{\ominus} \end{matrix} \leftrightarrow CH = CH - \overset{\oplus}{N} \begin{matrix} O^{\ominus} \\ // \\ \diagdown O^{\ominus} \end{matrix}$ (I) (II) (I > II)
Answer	2

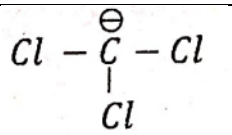
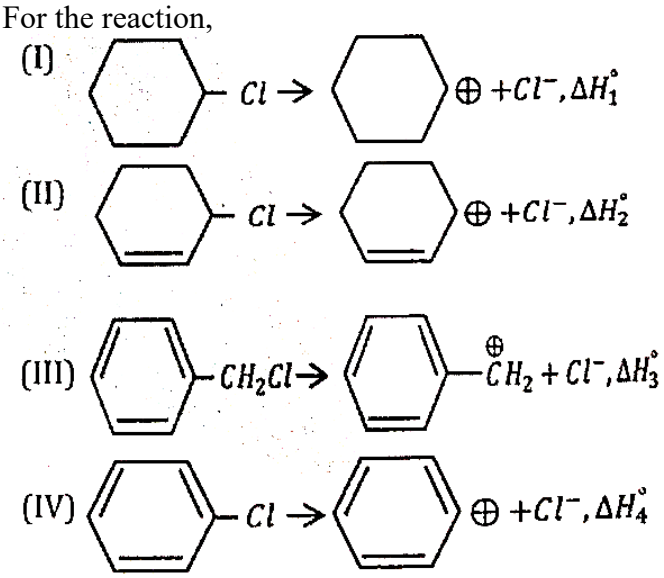
Solution	
Positive Marks	4
Negative Marks	1
Question	Which order is true for resonance energy?
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In which of the following compounds lone pair indicated is involved in resonance?
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following species does not show hyperconjugation?
Type	multiple choice
Option	$C_6H_5-CH_3$

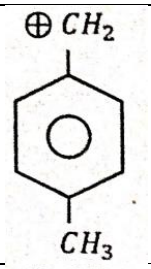
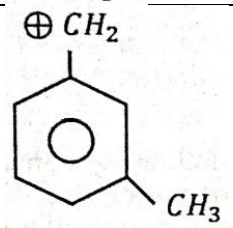
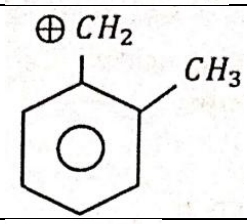
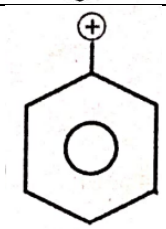
Option	
Option	
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following alkenes will show maximum number of hyperconjugation forms?
Type	multiple choice
Option	
Option	$CH_3 - CH = CH - CH_3$
Option	
Option	
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following molecules has longest C = C bond length?
Type	multiple choice
Option	$CH_3 - CG = CH - CH = CH - CH_3$
Option	$CH_2 - CH - CH = CH_2$
Option	$CH_3 - CH = CH - CH_3$
Option	$CH_2 = CH_2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is not an aromatic compound?
Type	multiple choice

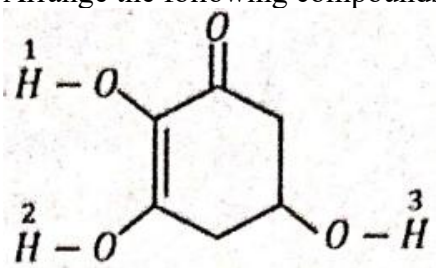
Option	
Option	
Option	
Option	
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Identify the correct statement (s): 
Type	multiple choice
Option	I and III are antiaromatic
Option	III and IV are non-aromatic
Option	II and III is aromatic
Option	II and IV are non-aromatic
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following cations is hyperconjugation destabilized?
Type	multiple choice
Option	$CH_3-\overset{\oplus}{C}H_2$
Option	$CH_3-\overset{\oplus}{C}H(CH_3)$
Option	$CH_3-C^{\oplus}(CH_3)_2$
Option	
Answer	4

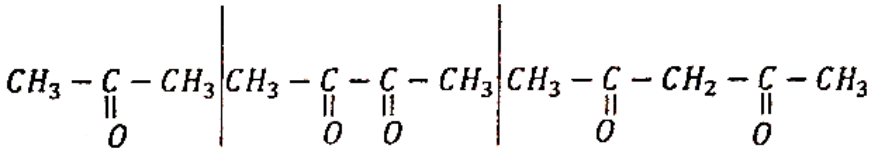
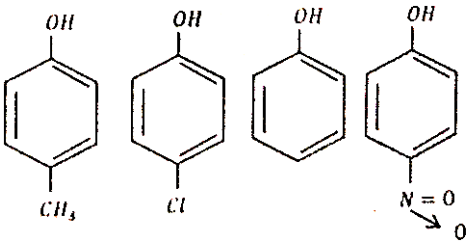
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following alkenes is the most stabilized?
Type	multiple choice
Option	$CH_2 = CH_2$
Option	
Option	
Option	
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following alkane have maximum heat of combustion
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Choose the correct order of heat of combustion? 
Type	multiple choice
Option	2 > 1 > 3 > 4
Option	1 > 4 > 2 > 3
Option	4 > 3 > 2 > 1
Option	4 > 1 > 2 > 3
Answer	3

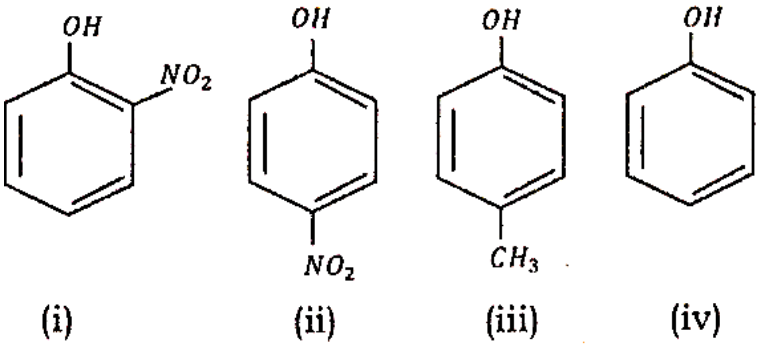
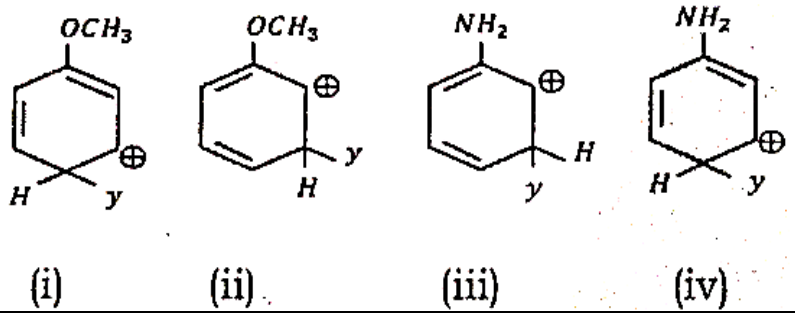
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following has least heat of hydrogenation?
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is correct decreasing order of bond dissociation energies of the bonds indicated with the arrows? 
Type	multiple choice
Option	c > a > b
Option	c > a > b
Option	c > b > a
Option	b > c > a
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is the most stable?
Type	multiple choice
Option	$C^{\ominus}H_3$
Option	$CH_3-\overset{\ominus}{C}(CH_3)-CH_3$
Option	$F-\overset{\ominus}{C}(F)-F$

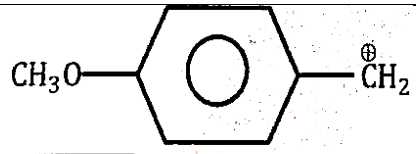
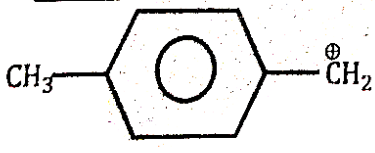
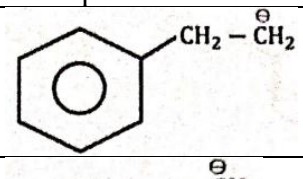
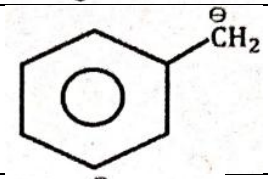
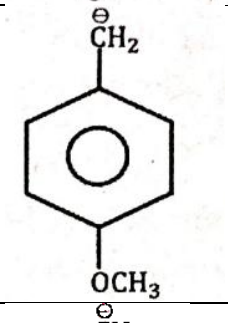
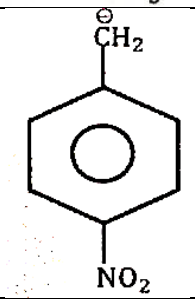
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following species is the most stable?
Type	multiple choice
Option	$p-O_2N-C_6H_4-C^+H_2$
Option	$C_6H_5-CH_2^+$
Option	$p-Cl-C_6H_4-C^+H_2$
Option	$p-CH_3O-C_6H_4-C^+H_2$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reaction,  The correct decreasing order of enthalpies of reaction for producing carbocation is
Type	multiple choice
Option	$\Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ > \Delta H_4^\circ$
Option	$\Delta H_4^\circ > \Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ$
Option	$\Delta H_3^\circ > \Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ$
Option	$\Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ > \Delta H_3^\circ$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is the most stable carbocation?
Type	multiple choice

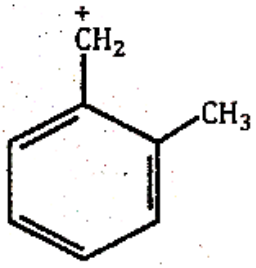
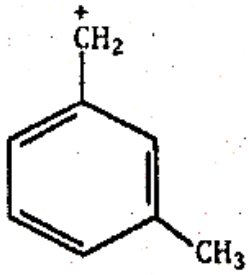
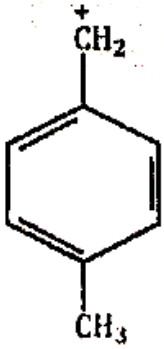
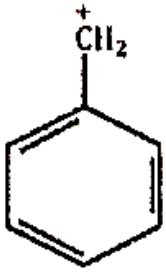
Option	
Option	
Option	
Option	
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For the following acids $(CH_3)_3CCH_2CO_2H$ (I) $(CH_3)_3\overset{\oplus}{N}CH_2CO_2H$ (II) CH_3COOH (III) pK_a value will be in order
Type	multiple choice
Option	I > II > III
Option	I >II > III
Option	I > III > II
Option	II > III > I
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	pK_a value of the compound decreases if X is
Type	multiple choice
Option	$-NO_2$
Option	$-NH_2$

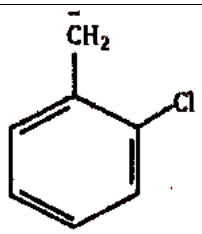
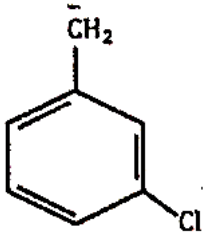
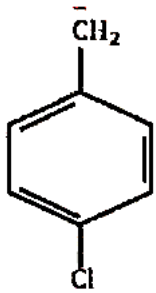
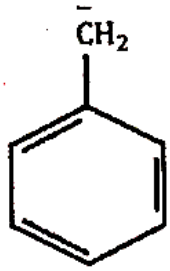
Option	$-OH$
Option	$-OCH_3$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following arrangements is in correct order of acidic strength of the given compounds? (I) CH_3-NO_2 (II) $NO_2-CH_2-NO_2$ (III) $CH_3-CH_2-NO_2$ (IV) $NO_2-CH(NO_2)-NO_2$
Type	multiple choice
Option	IV > II > I > III
Option	IV > II > III > I
Option	III > I > II > IV
Option	III > I > IV > II
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following options is correct for order of acidity of the compounds given? (I) $CH\equiv C-COOH$, (II) $CH_2=CH-COOH$, (III) CH_3CH_2COOH
Type	multiple choice
Option	I > II > III
Option	III > II > I
Option	II > I > III
Option	I > III > II
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Arrange the following compounds for their acidic strength order. 
Type	multiple choice
Option	2 > 1 > 3

Option	1 > 2 > 3
Option	1 > 3 > 2
Option	2 > 3>1
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Arrange the following compounds for their acidic strength order.  (I) (II) (III)
Type	multiple choice
Option	$I > II > III$
Option	$III > II > I$
Option	$I > III > II$
Option	$III > I > II$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Among the following compounds, the strongest acid is
Type	multiple choice
Option	$HC \equiv CH$
Option	NH_3
Option	$CH_2 = CH_2$
Option	CH_3OH
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following shows the correct order of decreasing acidity?
Type	multiple choice
Option	$PhCO_2H > PhSO_3H > PhCH_2OH > PhOH$
Option	$PhSO_3H > PhOH > PhCO_2H > PhCH_2OH$
Option	$PhCO_2H > PhOH > PhCH_2OH > PhSO_3H$
Option	$PhSO_3H > PhCO_2H > PhOH > PhCH_2OH$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the correct order of Acidic Strength?  (i) (ii) (iii) (iv)

Type	multiple choice
Option	$iv > iii > ii > i$
Option	$iv > ii > iii > i$
Option	$iv > ii > i > iii$
Option	$ii > i > iii > iv$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the correct order of Acidic Strength? 
Type	multiple choice
Option	$i > ii > iv > iii$
Option	$ii > i > iv > iii$
Option	$iii > ii > i > iv$
Option	$ii > iv > iii > i$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is the correct stability order of the given compounds? 
Type	multiple choice
Option	$I > II > III > IV$
Option	$IV > I > II > III$
Option	$IV > I > III > II$
Option	$IV > II > I > III$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Consider the following carbocations

	(1)  (2) (3)  (4) $CH_3-\overset{\oplus}{C}H_2$ The relative stabilities of these carbocations are such
Type	multiple choice
Option	$4<2<3<1$
Option	$2<4<3<1$
Option	$4<2<1<3$
Option	$2<4<1<3$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The most stable carbanion among the following is
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1

Question	For carbocation, (1)  (2)  (3)  (4)  Stability order will be:
Type	multiple choice
Option	$1>2>3>4$
Option	$1>2>4>3$
Option	$1>3>2>4$
Option	$4>3>2>1$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For carboanion,

	(1)  (2)  (3)  (4)  Stability order will be:
Type	multiple choice
Option	$1 > 2 > 3 > 4$
Option	$1 > 2 > 4 > 3$
Option	$1 > 3 > 2 > 4$
Option	$4 > 3 > 2 > 1$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1